

Pathology-Level ML Model Performance Summary (Test Set) Classical ML models trained on neural network pathology predictions							
Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Mean Pool	A Lines	Random Forest	0.687	[0.619-0.754]	43689	0.472	5
Mean Pool	A Lines	Logistic Regression	0.741	[0.694-0.787]	43689	0.472	5
Mean Pool	A Lines	XGBoost	0.733	[0.686-0.781]	43689	0.472	5
Mean Pool	Large Consolidation	Random Forest	0.712	[0.609-0.815]	43689	0.110	5
Mean Pool	Large Consolidation	Logistic Regression	0.831	[0.740-0.921]	43689	0.110	5
Mean Pool	Large Consolidation	XGBoost	0.776	[0.673-0.880]	43689	0.110	5
Mean Pool	Pleural Effusion	Random Forest	0.601	[0.515-0.688]	43689	0.011	5
Mean Pool	Pleural Effusion	Logistic Regression	0.714	[0.608-0.819]	43689	0.011	5
Mean Pool	Pleural Effusion	XGBoost	0.744	[0.629-0.860]	43689	0.011	5
Mean Pool	Other Pathology	Random Forest	0.558	[0.526-0.589]	43689	0.405	5
Mean Pool	Other Pathology	Logistic Regression	0.609	[0.602-0.616]	43689	0.405	5
Mean Pool	Other Pathology	XGBoost	0.588	[0.531-0.644]	43689	0.405	5
Inception3D	A Lines	Random Forest	0.519	[0.472-0.565]	43689	0.472	5
Inception3D	A Lines	Logistic Regression	0.584	[0.572-0.595]	43689	0.472	5
Inception3D	A Lines	XGBoost	0.552	[0.517-0.586]	43689	0.472	5
Inception3D	Large Consolidation	Random Forest	0.527	[0.494-0.561]	43689	0.110	5
Inception3D	Large Consolidation	Logistic Regression	0.550	[0.526-0.573]	43689	0.110	5
Inception3D	Large Consolidation	XGBoost	0.558	[0.533-0.584]	43689	0.110	5
Inception3D	Pleural Effusion	Random Forest	0.622	[0.552-0.691]	43689	0.011	5
Inception3D	Pleural Effusion	Logistic Regression	0.643	[0.612-0.675]	43689	0.011	5
Inception3D	Pleural Effusion	XGBoost	0.717	[0.666-0.769]	43689	0.011	5
Inception3D	Other Pathology	Random Forest	0.534	[0.498-0.570]	43689	0.405	5
Inception3D	Other Pathology	Logistic Regression	0.604	[0.600-0.607]	43689	0.405	5
Inception3D	Other Pathology	XGBoost	0.566	[0.535-0.597]	43689	0.405	5
Levit-Attention	A Lines	Random Forest	0.548	[0.535-0.560]	43689	0.472	5
Levit-Attention	A Lines	Logistic Regression	0.586	[0.581-0.592]	43689	0.472	5
Levit-Attention	A Lines	XGBoost	0.564	[0.553-0.576]	43689	0.472	5
Levit-Attention	Large Consolidation	Random Forest	0.517	[0.497-0.537]	43689	0.110	5
Levit-Attention	Large Consolidation	Logistic Regression	0.543	[0.492-0.593]	43689	0.110	5
Levit-Attention	Large Consolidation	XGBoost	0.530	[0.509-0.551]	43689	0.110	5
Levit-Attention	Pleural Effusion	Random Forest	0.590	[0.541-0.639]	43689	0.011	5
Levit-Attention	Pleural Effusion	Logistic Regression	0.655	[0.624-0.687]	43689	0.011	5
Levit-Attention	Pleural Effusion	XGBoost	0.693	[0.616-0.769]	43689	0.011	5
Levit-Attention	Other Pathology	Random Forest	0.562	[0.543-0.582]	43689	0.405	5
Levit-Attention	Other Pathology	Logistic Regression	0.606	[0.602-0.611]	43689	0.405	5
Levit-Attention	Other Pathology	XGBoost	0.579	[0.548-0.610]	43689	0.405	5